Roll No……..

EVEN END Semester Examinations June 2025

Name of the Program: **B. Tech (ECE-ESR)** *Semester: IV*
Name of the Course: **Introduction to Robotics** *Course Code:* **TEC 451**

Time: 3 Hours **Maximum Marks: 100**

Note:

(i) All questions are compulsory.
(ii) Answer any two sub-questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	**CO1, CO3**
(a)	How do end effectors influence the precision of movement in robotic systems, and what design considerations are critical to maintaining accuracy during task execution?	
(b)	Briefly describe the basic working principle of each robot drive system with a supporting block diagram?	
(c)	How are transducers used in conjunction with sensors to enable real-time feedback and decision-making in robotic systems, and what are their applications in industrial and service robotics?	
Q2	(20 marks)	**CO2, CO3**
(a)	How do controllers, sensors, and actuators work together in a robotic system, and what is the specific role of velocity sensors in ensuring accurate motion control?	
(b)	What are the potential challenges and limitations in applying Asimov's Three Laws of Robotics to real-world robots, particularly in dynamic environments like healthcare or autonomous vehicles?	
(c)	How does the derivative component impact the transient response of a control system, particularly in reducing overshoot and settling time?	
Q3	(20 marks)	**CO3, CO5**
(a)	In what ways is the future of the robotics market being transformed by AI technologies such as machine learning, computer vision, and autonomous decision-making?	
(b)	How do power transmission systems affect the performance, efficiency, and precision of robotic movements, particularly in complex robotic arms or mobile robots?	
(c)	What are the key characteristics of a sensor that impact its effectiveness in machine vision, such as resolution, sensitivity, and field of view?	
Q4	(20 marks)	**CO3, CO4**
(a)	Given a vector v=3i+4j−k, it rotates by 60° about the z-axis of the reference frame. What is the rotation transformation for this vector?	
(b)	What is the difference between forward kinematics and inverse kinematics in robotic manipulators, and why are both important for motion planning?	
(c)	For the vector v=4i−3j+5k, perform a translation by a distance of 3 meters in the x-direction, 6 meters in the y-direction, and 2 meters in the z-direction. What is the resulting translation transformation?	
Q5	(20 marks)	**CO5**
	What are the key components of a machine vision system in robotics,	

(a)	and how do cameras, sensors, and processing units work together to capture and interpret visual data?	
(b)	A 7 kg rectangular block is gripped by a two-finger gripper and lifted vertically at a velocity of 2 m/s. If the acceleration is 15 m/s^2 and the coefficient of friction between the gripping pads and the block is 0.35, calculate the minimum force required to prevent slippage.	**CO6**
(c)	What are the key image processing techniques, such as edge detection, filtering, and segmentation, and how do they contribute to a robot's vision capabilities?	